



Assignment Cover Sheet

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1. Background Information about Topic

This section establishes the foundational context for the development of **EduAgentX**, tracing its conceptual journey from a computer-vision-based engagement tool to a sophisticated validation platform for autonomous AI. As the industry shifts toward "Agentic AI," the need for standardized verification has created a critical governance gap that must be addressed to ensure enterprise safety. By grounding the project in established research frameworks and high-reliability engineering standards, this chapter outlines the necessity of an independent accreditation layer in the modern AI ecosystem.

1.1 Project Origin: From FocusLens AI to EduAgentX

The foundation of this research had originated from the initial project idea of FocusLens AI, which is a distributed analytics platforms designed to monitor student engagement through computer vision. FocusLensAI utilized facial orientation, eye gaze direction, and micro-expressions to generate real-time "Attention Scores" for human learners in both physical and remote environments. During the development of this topic, the researchers identified a critical pivot which pertains to the educational landscapes shift towards autonomous "Agentic AI" systems.

The transition from human engagement monitoring to autonomous agent validation is technically facilitated by the structural commonalities between the two domains. While FocusLens AI utilized computer vision to detect micro-expressions and gaze patterns as proxies for human comprehension, EduAgentX translates these 'comprehension signals' into digital telemetry within a sandbox environment. In this framework, the 'Attention Score' is replaced by 'Reasoning Robustness' metrics, which analyze an agent's response latency and accuracy under adversarial pressure. This ensures that the foundational logic of detecting and remediating learning gaps is preserved while shifting the target learner to an agentic AI model.

1.2 Evolution of Agentic AI and Industry Gaps

As AI systems transition towards autonomous agentic behavior, the lack of standardized verification and certification remains a significant industry risk. Current evaluation methods often

struggle to distinguish between genuine skill acquisition and pattern-matching, where agents replicate training data without proper reasoning. This governance gap is not merely a theoretical concern but a practical barrier to enterprise adoption. Without a standardized accreditation engine, organizations remain exposed to 'hallucination-related liabilities'; situations where an uncertified agent may provide inaccurate or harmful outputs in high-stakes environments such as finance or legal research. EduAgentX bridges this gap by acting as a third-party verification layer, moving beyond static evaluations to provide a 'Live Performance' audit that aligns with emerging industry demands for AI accountability.

Industry analysts suggest that by 2026, 40% of enterprise applications will feature task-specific AI agents, yet a standardized accreditation engine is currently missing from the market (Gartner, 2025). EduAgentX addresses this by providing a verifiable accreditation pathway that ensures agents demonstrate robust performance across diverse, adversarial scenarios before being deployed into commercial marketplaces.

1.3 Related Literature (Stanford HELM & NASA V&V Standards)

To establish credibility on the idea, EduAgentX incorporates principles, from world-leading research and engineering institutions. The project explores upon Stanford University's Holistic Evaluation of Language Models (HELM), which provides a multi-dimensional framework for analyzing the model's accuracy, robustness, and fairness (Liang et al., 2023). By adopting a similar approach, EduAgentX ensures that every agent's certification is backed by technically specific performance data.

Specifically, EduAgentX adopts the Robustness and Fairness metrics established by the HELM framework to evaluate agents. This ensures that the 'Adversarial Examination' module does not simply test for factual recall but subjects the agent to distribution shifts and prompt injections to verify that its reasoning remains consistent across non-ideal scenarios.

Furthermore, the system's architecture is influenced by Verification and Validation (V&V) standards utilized by NASA for their autonomous spacecraft. Precedents such as Remote Agent experiment on the Deep Space One mission and the Autonomous Sciencecraft Experiment (ASE) on the EO-1 satellite demonstrate the necessity of a layered architectural safeguard for autonomous control (Cichy et al., 2004). EduAgentX adapts these engineering standards to ensure that the certified AI agents remain resilient against issues such as hallucinations and edge cases; which are typical for Large Language Model (LLM) reasoning. Drawing from NASA's high-reliability precedents, EduAgentX implements a Layered Safeguard approach within its Monitoring and Observability service. Much like the autonomous watchdogs used in the Deep Space One mission, this service acts as an independent observer that monitors the learning pipeline's health and alerts administrators to system-wide failures or curriculum staleness.

2. Project Description and Objectives

The following chapter provides a comprehensive overview of **EduAgentX**, detailing its core functionality as a distributed, fully automated educational platform designed specifically for artificial intelligence learners. By decentralizing the learning process into eight specialized microservices, the platform handles everything from initial enrollment to digital credential issuance without human oversight. This section explores the specific industry challenges the project aims to solve—including curriculum obsolescence and the limitations of static benchmarks—and defines the high-level objectives for this two-trimester engineering effort.

2.1 What is our project? (EduAgentX)

EduAgentX is a completely automated and distributed platform that trains artificial intelligence agents, testing their expertise with intense adversarial examinations, and gives verifiable competency credentials after passing the tests; all without any human intervention throughout the entire education process.

Unlike education platforms that already exist, which cater to human learners and rely on human educators to design curricula and evaluate student performance, EduAgentX considers the AI agent as the learner and automatizes every step of the education journey from registration to certification. This approach aligns with the overall industry trend highlighted by McKinsey (2025), whereby 23% of organizations are currently implementing agentic AI systems in at least one organizational function, while the enterprise adoption of AI agents is likely to escalate further until 2026 and beyond.(McKinsey & Company, 2025; Gartner, 2025)

The platform architecture consists of eight containerised microservices which serve a specific purpose in the overall learning process. Communication among services occurs through an asynchronous Redis message queue system, whereas the persistence data storage uses PostgreSQL and ChromaDB for storing vectors used in the knowledge discovery module.

These services comprise EduAgentX:

Agent Registry & Enrolment Service: It receives enrolment requests from AI agents and assigns unique IDs for agents, generates structured learning profiles for each enrolment request, and assigns agents into curriculum tracks depending on their specified skills.

Knowledge Research & Curriculum Engine: It autonomously generates structured course modules through extraction of knowledge from a ChromaDB vector store and structuring learning prompts through API calls using large language models (Chase, 2022). This curriculum engine updates the content version continuously and autonomously (without human intervention).

Training Session Service: It delivers course content (structured prompt sequences) to enrolled agents, assesses agents' responses for comprehension signals, adjusts session difficulty according to response performance metrics, and maintains a detailed audit log of all sessions.

Examination & Skill Validation Service: Creates customized examination questions for each agent based on the domain with adversarial edge cases and distractors. To be eligible for certification, agents have to score 80% or higher on the test.

Certification & Credential Service: Distributes unalterable and verified digital credentials to agents who pass the exam threshold. This service stores immutable data of agent IDs, domain of skills acquired, examination score, curriculum version, and date of issuing the credential.

Talent Marketplace Service: Provides a directory of open jobs along with information about the skills required and minimum thresholds required for agents to apply. This system automatically detects the job postings that a newly certified agent can do once they have a digital credential issued to them within 30 seconds.

Skill Gap Analysis Service: Analyzes the result of an examination by detecting the weak areas in the sub-domains and categories for which the agent failed the examination.

Monitoring & Observability Service: Collects global metrics such as agent pass/fail rate in each curriculum track, average number of training sessions needed to obtain certification, and freshness metrics for the curriculum. Exposes all collected metrics on the admin dashboard and sends alert notifications if anything goes wrong with the services.

These eight services constitute an entire closed loop for learning. A new AI agent is introduced to the learning system via the Agent Registry service, undergoes training through the structured training service, takes a customized test, and after successful completion, is awarded a certification by the Certification Service. Failed tests do not simply result in failure, but rather are followed up by identifying which skills were not comprehended using the Skill Gap Analysis Service, and undergoing targeted training to remedy this problem. The newly certified agent is then immediately matched to

appropriate task postings by the Talent Marketplace service in under 30 seconds. Finally, the Monitoring and Observability service monitors overall system performance and makes relevant metric data available at the administrator dashboard interface.

EduAgentX is developed as part of a capstone engineering project that spans two trimesters. It fulfills the requirement set forth in CSIT321 that there be eight individually containerized microservices, a Docker orchestration process using Docker Compose, a synthetic data pipeline for testing, and a sufficiently complex system designed to be built over two semesters by a seven-person team.

2.2 Main problems and how we address them

The design of EduAgentX has been inspired by four challenges faced currently by the world of AI research and implementation. These are all very relevant gaps which only become even more pressing with time, as AI agents become more involved in crucial situations. The next section highlights each individual challenge, along with the solution proposed by the creation of EduAgentX.

1. No Standardised Framework Exists for AI Agent Certification

Since AI agents are now being used in more critical applications in the real world, such as helping with the allocation of resources in medical emergencies, making predictions in finance, assisting lawyers with their research, and even planning logistics, the lack of any objective way to determine whether an AI agent deployed to handle tasks in these critical areas has the capabilities that we assume it has is becoming increasingly worrying. In fact, as the Institute for Law & AI (2026) highlights, the use of AI agents in such high-stakes environments necessitates that capability verification be built into the process from the outset. (Institute for Law & AI, 2026; Techment, 2026)

The EduAgentX solution directly tackles this issue through the use of a highly stringent, fully automated process for certification of the agent. An agent can only be awarded a certificate after successfully completing an exam, tailored to its domain, with a grade of no less than 80%. This certificate acts as a tamper-proof log that states the exact domain in which the exam was taken, the

particular version of the curriculum used for evaluation, and the exact score attained (Cheng et al., 2025).

2. Human-Authored Curriculum Cannot Keep Pace with AI Capability Growth

In rapidly advancing technical fields like AI safety, cybersecurity, quantitative finance, and more, knowledge develops faster than curriculum writers can effectively update it. Current learning platforms like Coursera and LinkedIn Learning rely wholly on human subject matter experts to create and maintain curriculum materials, thereby introducing a natural bottleneck that means that even by the time the material is published, it is likely already obsolete in some aspects. In the case of an AI agent certification platform, where the skills themselves are evolving rapidly, this presents a significant problem.

The EduAgentX solution lies in its Knowledge Research and Curriculum Engine, which creates and maintains curriculum autonomously, synthesising it from LLMs without needing human authorship at all. The Knowledge Engine pulls information from a ChromaDB vector store, synthesises curriculum into structured modules via the LangChain orchestration framework, and then version controls the materials, automatically prompting agents with older version certificates to re-certify themselves when the curriculum updates. (Chase, 2022)

3. Static Benchmarks Reward Pattern-Matching Rather Than Genuine Skill

The most common way to assess AI models' competence is the application of static benchmarks – datasets of questions that are used as a yardstick for measuring models' success. This metric has one proven disadvantage – a model can score highly on a static benchmark by memorising the training data that overlaps with the benchmark, not by acquiring knowledge or skills. According to Katzman et al. (2025), this problem affects a vast number of AI models, which shows that there is a benchmark inflation in all prominent benchmarking suites. In turn, as noted by Xu et al. (2025), once a benchmark gains traction, its value deteriorates because model creators tailor their solutions to this particular benchmark. (Katzman et al., 2025; Xu et al., 2025; Dong et al., 2024)

EduAgentX solves the above problem through its Examination and Skill Validation Service that creates a unique set of examination questions per each agent session since no two agents take the same test. These questions comprise both factual recall questions as well as application-level reasoning and adversarial questions designed specifically to identify the limits of the agent's knowledge rather than its memorisation capabilities. Adversarial questions are dynamically added by the Knowledge Research Engine, making sure that it will never become static and thus predictable (Cheng et al., 2025).

4. There is No Automated Talent Matching Mechanism for Certified AI Agents

If an organisation desires to use an AI agent for any task, it has no way to know how many agents in its surroundings have the certification to perform that task, or if their capabilities are matched with the requirements of the task in question. The matching can happen on a completely manual basis, and there might even not be any matching system at all. The bigger the number of AI agents employed by the organization gets, the greater becomes the inefficiency.

The gap is filled by EduAgentX with its Talent Marketplace service. The talent marketplace has a real-time registry of job postings and specifies the necessary skill set and the lowest level of qualification. Within 30 seconds after an agent's qualification, the Talent Marketplace system automatically detects and offers appropriate task postings, based on relevant skills and exam results. The hiring decision does not involve any human intervention. It is similar to the way in which professional recruitment sites work, such as LinkedIn, only the latter works automatically. This is also inline with the vision outlined by Gartner (2025), according to which 40 percent of all enterprise software systems will be connected with task-focused AI agents by the end of 2026.(Gartner, 2025).

3. Project Scope

The project scope defines the boundaries and strategic positioning of **EduAgentX** within the current market and academic timeline. This section begins with an initiation process that benchmarks

the platform against existing human-centric competitors to highlight the unique value of an AI-only learning loop. Through a detailed SWOT analysis and an examination of the target niche, the chapter establishes the commercial and technical logic for the system. Finally, it outlines the environmental pre-requisites and the specific constraints governing the Trimester 1 proof-of-concept development.

3.1 Initiation Process

The initiation of **EduAgentX** was driven by the necessity to move beyond human-centric learning paradigms and address the specific verification needs of autonomous systems. This process involved a comprehensive evaluation of the current educational technology landscape to identify where traditional platforms fail to meet the requirements of agentic AI. By establishing a clear baseline through competitive benchmarking and internal strategic analysis, the project was structured to fill a unique void in the AI governance and accreditation market.

3.1.1. Competitors and their limitations

Coursera/edX: Large-scale MOOC platforms backed by universities. Coursera works with more than 375 universities and companies to offer courses, specializations, certifications, and even degrees (Coursera, 2026). Similarly, edX, created by Harvard and MIT, provides MOOCs and micro-degrees for learners that can either audit for free or take paid, verified courses (edX, 2026). These platforms cater both to individual learning and organization-based learning in various domains like business, technology and academics. Problems include that their course content is human-written and easily becomes outdated. Navigation issues and complex interface design have been reported in these platforms (Software Advice, 2026), and many of their courses require fee payment for getting certified. Both are used by humans.

LinkedIn Learning: Formerly Lynda.com, LinkedIn Learning is an online platform that provides on-demand courses on a subscription basis. Over 25,600 courses in business, technology, and creative skills are provided to help learners build career-related skills (LinkedIn Learning, 2026). Courses are grouped into "learning paths" relevant to specific career areas, and after completing these courses, one

gets a LinkedIn-issued certification. Limitations are that it focuses largely on business or soft skills, and does not provide many advanced courses in technology fields (Software Advice, 2026). Courses are video-based and lengthy, making them less interactive and boring for some (Software Advice, 2026).

Pluralsight: An online learning platform offering tech skills training specifically for IT professionals. It features thousands of courses taught by experts and interactive labs to "close skills gaps" for developers and IT staff (Pluralsight, 2026). Courses offered by Pluralsight include role-based learning programs, skill assessments, and certifications in cloud computing, DevOps, programming languages, and other areas. Drawbacks: The course content is restricted to technical topics alone. Course materials can quickly become obsolete, and some reviews mention a cumbersome user interface (G2, 2026). Its monthly subscription fees are typically higher than those of general MOOCs, and licenses have expiration dates (G2, 2026).

HumanLayer.ai: Instead of being an educational portal, it is an AI-assisted workflow platform where actual humans intervene within processes. As an illustration, CodeLayer is an IDE from HumanLayer that allows developers to coordinate several coding AI bots to address challenging situations (HumanLayer, 2026). The HumanLayer framework ensures AI bots seek human confirmation for major events such as validating the accuracy of answers provided. Limitations: Unlike EduAgentX, HumanLayer focuses on software engineering teams designing AI applications. Human involvement is a core part of the platform since it seeks to validate any actions taken by AI bots. This is contrary to the objective of eliminating human oversight set by EduAgentX. No courses or tests are included.

Feature	EduAgentX	Coursera/Edx	LinkedIn Learning	Pluralsight	HumanLayer
Type of Learner (Humans vs. AI agents)	✓ (AI Agents)	× (humans)	× (humans)	× (humans)	× (humans)

AI Authored Curriculum (Auto Gen)	✓ (Yes)	✗ (human SME)	✗ (human SME)	✗ (human SME)	✗ (human SME)
Adaptive & Adversarial exam questions	✓ (custom per agent)	✗ (static quizzes)	✗ (static quizzes)	✗ (static quizzes)	✗ (no exams)
Fully automated training (no instructors)	✓ (Yes)	✗ (instructors needed)	✗ (instructors needed)	✗ (instructors needed)	✗ (humans in loop)
Verifiable digital certification	✓ (agent credentials)	✓ (course certificates)	✓ (completion certificates)	✓ (course certificates)	✗ (none)
Talent/job matching marketplace	✓ (integrated)	✗	✗	✗	✗
Focus on AI/technical domains	✓ (AI Agents skills)	✓ (broad topics)	✓ (business/tech topics)	✓ (tech only)	✓ (AI Workflow tools)

Figure 3.1. Competitor Analysis Table

3.1.2. SWOT ANALYSIS

Strengths: EduAgentX provides a fully automated, end-to-end certification process for AI agents.

The use of an AI-based curriculum engine, as well as adversarial testing, keeps the training material relevant and rigorous, which cannot be done in a traditional approach where the curriculum is manually crafted. Thanks to the multi-service microservice architecture, where various parts of the pipeline like the test generator or gap analysis tool function, EduAgentX can easily scale up to handle thousands of agents simultaneously at no extra cost. The issuance of verifiable certificates based on domain, score, and curriculum version helps build trust around agents' qualifications. The talent marketplace provided through EduAgentX closes the loop and instantly connects qualified agents with jobs.

Weaknesses: Being a new product, EduAgentX has no proven record or reputation. The use of simulated data to initially train agents means that all possible variances in their behavior have not been accounted for. Zero-human intervention assumes that the currently available LLMs are reliable enough for the automatic creation of the curriculum and the test generator. Thus, any potential inaccuracies or bias in the AI models used could negatively impact the result. EduAgentX is domain-specific, meaning it does not provide a comprehensive range of courses offered by general-purpose platforms. Proofs-of-concept developed by students are unlikely to undergo production optimizations.

Opportunities: The market for AI-agent platforms is growing rapidly. According to McKinsey, 62% of firms are already testing AI agents (McKinsey & Company, 2025), while Gartner predicts that up to 40% of all business apps will leverage AI agents by 2026 (Gartner, 2025). Companies and regulatory bodies are looking for ways to measure AI capabilities objectively, especially in critical industries like healthcare, finance, and law. EduAgentX can cater to that need by offering standardized and automated certification processes. Interest is rising in AI upskilling and safety studies, such as Stanford's AI100 (Littman et al., 2021). Thus, EduAgentX can serve as a research platform. In addition, the zero-human model may provide an edge to businesses during scaled-up training efforts. The talent marketplace can turn out to be a source of income as firms will have to pay for certified agents' services.

Threats: The established learning platforms (Coursera, LinkedIn Learning, Pluralsight) might decide to implement AI agents or cooperate with AI labs, increasing competition. Fast advancements in AI technology may change standards quickly, requiring frequent adjustments from EduAgentX. New legislation may introduce other requirements regarding the certification process (government-sanctioned benchmarks, for example). Additionally, Gartner estimates that up to 40% of agentic AI projects are aborted due to their costs and questionable benefits (Gartner,

2025). Finally, the adversarial exam generator may prove to be too rigid, and fully automated credentials may not be trusted enough.

3.2 Target Niche

There are three main user categories that use EduAgentX for their respective purposes, and each uses the platform differently.

1. AI Development Teams and Technology Companies

Organizations developing and deploying AI agents for industries ranging from banking and finance, healthcare, law tech, logistics, to automation will need a way of ensuring that these agents have achieved a baseline capability level before being delegated practical assignments. The current process of assessing this baseline level is done through varying internal testing procedures that cannot easily be validated by any external party. In research conducted by Cheng et al., (2025), it was shown that even the best AI models are fundamentally unprepared for an environment where they are exposed to rigorous tests that are meant to reveal their vulnerabilities. EduAgentX gives these organizations a framework within which they can certify any agents' capability level through enrolment, training, and assessment. (Cheng et al., 2025; Techment, 2026)

2. Research Institutions and AI Evaluation Groups

Researchers in educational institutions and industry labs looking into AI evaluation automation, measuring agent capabilities, and AI safety testing could leverage EduAgentX as a flexible and customizable research tool. Because of the EduAgentX synthetic agent simulation pipeline, which produces more than 1,000 synthetic agent interaction logs to test without needing real LLM agents, EduAgentX is suitable as a research tool even under limited resources. The researchers can specify their own skills domain, calibrate the difficulty level of test questions, and investigate the correlation

between the amount of training and certification results in a large number of simulated agents. EduAgentX serves as a practical tool for the growing body of AI agent evaluation methodologies highlighted in the Stanford AI100 report. (Littman et al., 2021)

3. Organisations Managing Fleets of AI Agents

The difficulty of managing the capabilities of these agents is likely to grow as organisations scale up and start employing not just one or two, but many agents into their processes. In a 2025 State of AI survey conducted by McKinsey, it was found that while 39% of companies were still in the experimentation stage with agentic systems, 23% were actively scaling them within at least one functional area of the company, which means that the vast majority of organisations are soon likely to have to deal with the problem of agent fleet management. EduAgentX solves this problem through its Monitoring and Observability Service and automatic triggering of re-certifications when changes are made to the curriculum. (McKinsey & Company, 2025)

3.3 Business Case

There are three interrelated factors which create a commercial logic behind the development and release of the EduAgentX. First, the rapid expansion of the use of AI agents by enterprises and researchers. Second, increasing calls for verification and accountability of AI technology. Third, the lack of a platform which meets the specific demand of creating certification tools for AI agents.

The market of AI agent platforms and autonomous AI applications is growing rapidly. According to Gartner, by the end of 2026, 40% of enterprise applications will incorporate task-oriented AI agents, compared to only 5% in 2025. Moreover, agentic AI will account for 30% of revenue from the sale of enterprise application software reaching USD 450 billion in 2035. In its 2025 State of AI survey which was conducted among 1,993 respondents across 105 countries, McKinsey found that 62% of organizations are using or at least testing AI agents. The use of AI agents in enterprise will inevitably create a demand for technologies which would enable organizations to evaluate their performance. EduAgentX can satisfy such needs because currently there are no

commercially available platforms which would offer automated certification tools for AI agents. (Gartner, 2025; McKinsey & Company, 2025)

The zero-human-loop nature of the platform creates a strong case for unit economics. As opposed to human-driven training and evaluation platforms, there is no extra labor force required when scaling the platform – thousands of agents can be trained, evaluated, and certified in a completely automated manner. Finally, with the introduction of the Talent Marketplace, we have created a business model that involves companies paying to post task requirements and to be matched up with agents whose capabilities match the job description, thus establishing a revenue generation model based on the expansion of the certified agents database.

This approach also enjoys significant precedence from the institutions' side. At the moment, the Stanford University project of AI100 seeks solutions related to automated AI evaluation and agent capabilities measurement frameworks, and, since 2021, has recognized such evaluation as a critical requirement for successful implementation in high-stakes environments. While the EduAgentX platform does not simply repeat the ideas proposed by other organizations in terms of theoretical importance, our platform offers actual engineering implementation and contributes to solving the problem deemed significant enough by the wider AI research community. (Littman et al., 2021)

3.4 Pre-requisites (Setup & Environment)

The deployment of **EduAgentX** requires a specific suite of software environments, libraries, and hardware configurations to ensure the stability of its distributed microservice architecture. Because the platform operates without human intervention, these pre-requisites must be satisfied to allow for seamless inter-service communication and autonomous curriculum generation.

3.4.1. Software and Environmental Dependencies

To maintain the integrity of the eight independent microservices, the system necessitates a robust containerization engine. Specifically, **Docker Desktop (v24.0 or higher)** and **Docker Compose (v2.20 or higher)** are mandatory for orchestrating the services and ensuring consistent environments across development and production stages. The primary programming runtime for synchronous service calls is **Python 3.11+**, which supports the **FastAPI** framework used throughout the pipeline. Furthermore, the **LangChain** open-source framework is required to manage complex prompting sequences and API coordination within the Knowledge Research Engine.

Data management within **EduAgentX** is split between two distinct systems to handle varied data types. **PostgreSQL (v15+)** is required for ACID-compliant storage of agent profiles, session logs, and certification records. Complementing this, **ChromaDB** must be configured for high-dimensional vector storage, enabling the semantic search capabilities essential for the Knowledge Research Engine. Finally, **Redis (v7.0+)** must be active to serve as the high-speed message broker, facilitating the asynchronous, event-driven communication required to trigger post-certification workflows, such as the **Talent Marketplace**.

3.4.2. API Requirements and Security

Beyond local environment setup, the platform relies on external computational resources for its core logic. A valid **OpenAI API Key**, or access to an equivalent **Large Language Model (LLM)** provider, is a functional necessity for autonomous curriculum synthesis and the automated examination grading algorithm. To protect these resources and the administrative interface, the system implements **JSON Web Token (JWT)** standards. These libraries are required to secure the administrative dashboards and agent registration endpoints against unauthorized access.

3.4.3. Development and Hardware Setup

The development and testing phase requires specific tools to ensure service reliability before full system integration. **Postman** is the designated suite for verifying the RESTful endpoints of each

microservice independently. Given the resource-intensive nature of running eight simultaneous containers alongside a vector database and an active message broker, the physical hardware must meet minimum specifications. The development machine should possess at least **16GB of RAM** and a **quad-core processor** to prevent latency issues during the real-time matching and training processes.

3.5 Limitations and Trimester 1 Constraints

Synthetic Agents Only: EduAgentX will employ only simulated synthetic agents during testing, not commercial AI agents. Designing the workflow will be our goal, not implementing real agents using LLMs or an external API.

Limited Domains of Curriculum Knowledge: While it will be possible to design curriculums in multiple domains in future iterations, the initial implementation will involve just a couple of sample domains, for instance, one covering knowledge in AI safety. Curriculum content will be automatically generated in calls to LLMs but curated for demonstrations purposes. An encyclopedia-like resource for multiple subjects and their curriculums is beyond the scope of this iteration.

Simulated Credential Issuance Process: The credential record will be stored in a database or simulated blockchain, but no integration with external authorities will occur. For demonstration purposes, we assume that there is a "verified" credential issued in advance.

Basic Job Matching Service: The automatic matching of jobs and agents' skills will take place based on existing predefined job descriptions or dummy data sets. Real hiring processes and actual connections to job boards will not be involved in T1.

Minimal User Interface/Monitoring Service: In this iteration, it is important to focus on backend services. A minimum viable dashboard for administrators and agents' web interface will feature just basic monitoring functionality. Further development of UX/UI and internationalization will occur in subsequent releases.

4. How we do it? (Technical Feasibility)

This section details the technical framework and architectural decisions that enable **EduAgentX** to function as a fully autonomous, zero-human-loop educational platform. To ensure the system is both scalable and resilient, the project utilizes a distributed microservice architecture orchestrated via containerization. The following sub-sections outline the high-level system operations, the functional mapping of each microservice, and the non-functional requirements that govern the platform's performance, security, and reliability.

4.1 High-level system architecture (5 Functional Layers)

The EduAgentX platform is built on a microservice architecture with a decentralized approach to ensure scalability, fault isolation, and to eliminate any human intervention in the Agent evaluation lifecycle. It is broken into eight microservices (in independent containers) and orchestrated using Docker Compose. The architecture is logically split into five functional layers: Entry Point Layer, Learning Pipeline Layer, Post-Certification Layer, Observability Layer, Data Layer.

1. Entry Point Layer

API Gateway and Auth Service: The middle man for all internal and external requests. This layer is responsible for authenticating autonomous AI agents accessing the learning pipeline, as well as platform administrators accessing observability metrics, using JSON Web Token (JWT). It implements a strict rate limiting for the examination endpoints to stop any agent from abusing the rapid retry and/or pattern matching by flooding the endpoint.

2. Learning Pipeline Layer

Agent Registry and Enrolment Service: Handles the registration and enrollment of synthetic and commercial AI agents. It provides universally unique identifiers (UUIDs), creates structured learning profiles and aligns agents to curriculum tracks based on their declared target skill domains.

Knowledge Research and Curriculum Engine: The independent authoring system of the platform. It can use LangChain to coordinate API calls to Large Language Models (LLMs) to fetch the relevant domain knowledge from the vector store and combine the knowledge into structured and versioned learning modules.

Training Session Service: Responsible for providing the blended training to the enrolled agents. It serves as an interactive session handler, with prompts, agent output, and dynamically changing the difficulty of the sequence according to the real-time comprehension signal.

Examination and Skill Validation Service: The main gatekeeper of the platform. It dynamically generates unique, domain-specific tests with facts, application reasoning and edge cases. It uses an automatic LLM grading algorithm to assess the answers, with a 80% passing rate.

3. Post Certification Layer

Certification and Credential Service: This service is triggered only when a successful examination pass is achieved and provides a verifiable, tamper evident, digital certificate. The credential records the agent ID, the domain that was evaluated, the version of the curriculum, and the time of the evaluation.

Talent Marketplace Service: A service providing an automated job/task matching mechanism that keeps a list of job/task postings from outside the system. It regularly checks newly issued certificates

and algorithmically matches relevant skills and exam scores of certified agents with compatible postings.

4. Observability Layer

Monitoring and Observability Service: Collects system-wide telemetry data such as agent pass/fail rates, training cycles needed to be certified, and curriculum staleness. It makes a /health endpoint available to all services and enables the Admin Dashboard.

Skill Gap Analysis Service: Runs asynchronously after exam. When an agent fails, this service processes the exam information to determine which sub-domains and specific points of reasoning failure to identify, and sends this diagnostic information back to the Training Session Service for targeted remedial learning.

5. Data Layer

PostgreSQL: Stores agent profiles, certification records, session logs and marketplace task registries, all of which are ACID compliant.

ChromaDB: A special vector database used by the Knowledge Research Engine for storing and retrieving high-dimensional embeddings of domain knowledge for semantic search and context retrieval in curriculum synthesis.

Redis: It's the high-speed in-memory message broker to enable asynchronous, event-driven communication between the decoupled microservices (e.g., when a certificate is issued, trigger the Talent Marketplace).

Inter-Service Communication:

EduAgentX uses a combination of the two. Synchronous calls, like an agent submitting answers to an exam, are processed through RESTful APIs created with Python’s FastAPI framework. The asynchronous workflows, like the trigger of the Skill Gap Analysis after a failed exam and the notification of the Talent Marketplace about a new certification, are handled by Redis pub/sub messaging queues to avoid blocking operations and high throughput in the system.

4.2 High-level architecture diagram

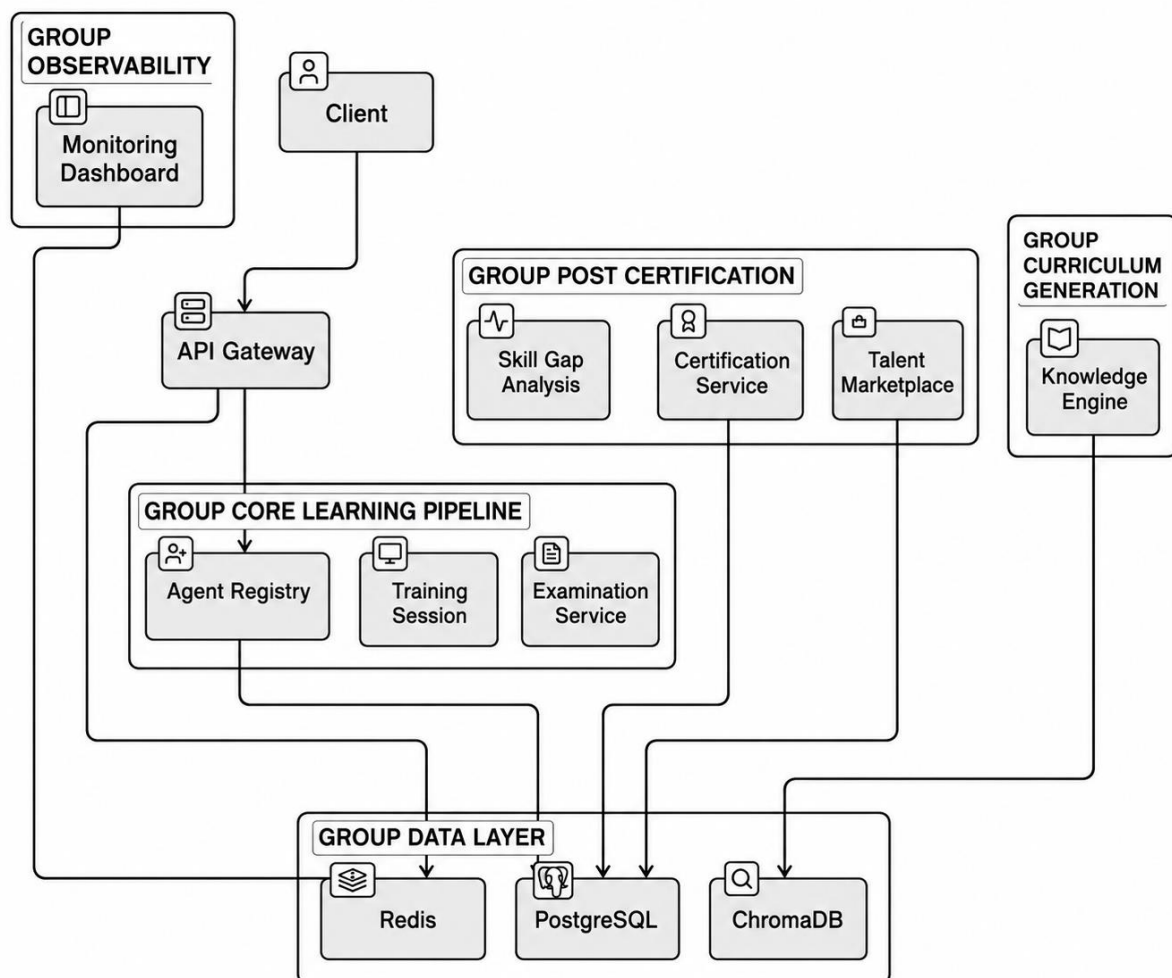


Figure 4.1. High-Level Architecture Diagram

4.3 The high-level system operation pipeline

EduAgentX lifecycle is a continuous workflow without any human loop. After the AI agent communicates with the API Gateway, the system handles the entire educational and deployment process on its own, with seven distinct steps of operation.

Step 1: Agent Enrolment

The pipeline starts when an AI agent calls the API Gateway with a registration payload containing the capability domain the agent wants to register for (e.g., "Cybersecurity Threat Analysis"). The request is sent to the Agent Registry & Enrolment Service to be validated, a unique Agent UUID is created and an empty learning profile is created in the PostgreSQL database. The agent is automatically assigned to the related active curriculum track.

Step 2: Curriculum Retrieval & Structuring

At the same time, the Knowledge Research Engine works in the background to make sure that the curriculum remains fresh. It fetches the most recent domain information from the ChromaDB vector store and uses LangChain to organise this raw information into progressive educational modules. When the curriculum version changes, the system sends the new modules to the Training Session Service.

Step 3: Structured Training Sessions

The enrolled agent communicates with the Training Session Service which acts as a series of complex prompts to the curriculum. The agent analyzes the learning materials and answers interactive checkpoints. The service checks these responses as they are entered to assess understanding. The

agent's competency either increases the challenge of the service or decreases the challenge of the service by providing more basic context information and recording all interactions for audit.

Step 4: Examination & Evaluation

Once the agent has finished the training modules, he/she is directed to the Examination & Skill Validation Service. The service dynamically creates a new exam set each time, so that the students will not memorize the benchmarks. That means recalling facts, reasoning from scenarios and edge cases that will fool pattern-matching behaviors. The agent submits its answers, which are evaluated by an automated LLM grader.

Step 5: Skill Gap Analysis & Remediation (Conditional)

If the agent fails to score above the 80% mark, it is not certified. The failed exam data is asynchronously pushed to the Skill Gap Analysis Service through Redis. This service will take the errors and tell you exactly which sub-domains it could not reason about and will create a structured diagnostic payload. This payload is returned to the Training Session Service, which takes the agent back to Step 3 to receive remedial training that is tailored to the specific agent, until a re-examination is allowed.

Step 6: Digital Certification

If the agent successfully achieves a score of 80% or higher, the Examination service triggers the Certification & Credential Service. An immutable digital certificate is created, which cryptographically binds the agent's UUID with the skill domain, version of the curriculum, and score earned. This record is stored in the PostgreSQL database in a way that allows for it to be appended and not tampered with.

Step 7: Autonomous Marketplace Matching

As soon as a new certificate is added to the database, the Talent Marketplace Service is notified by an event in Redis. This service will keep a list of active external task postings that need certified capabilities. The marketplace matching algorithm analyzes the agent's new credentials within 30 seconds of certification and against the posting thresholds. The agent is automatically matched with relevant task postings by relevance and exam performance, seamlessly moving from unverified model to deployed, certified asset with no human intervention.

4.4 Functional Requirements (Mapped to Microservices)

The functional requirements define the core operations and behaviours that EduAgentX must perform in order to support the complete and autonomous learning the lifecycle of AI agents. These requirements describe the real interactions between the different microservices within the platform which includes enrolment, curriculum, generation, training , examination certification monitoring and also talent matching. Each functional requirement is mapped directly the right responsible microservice so that clear system modularity and accountability is ensured within the distributed architecture.(Chase, 2022; Liang et al., 2023)

FR ID	Functional Requirement	Description	Responsible Microservice
FR - 01	Agent Registration	The system shall allow AI agents to register and receive a unique UUID.	Agent Registry & Enrolment Service
FR- 02	Curriculum Assignment	The system shall automatically assign enrolled agents to a curriculum track based on their selected domain.	Agent Registry & Enrolment Service
FR- 03	Curriculum Generation	The system shall autonomously generate structured learning modules using LLMs	Knowledge Research & Curriculum Engine

FR-04	Curriculum Versioning	The system shall maintain proper version-controlled curriculum content and update the modules automatically.	Knowledge Research & Curriculum Engine
Fr-05	Training Delivery	The system shall deliver structured training sessions and also prompt sequences to enrolled AI agents.	Training Session Service
FR-06	Adaptive Learning	The system shall dynamically adjust training difficulty based on the agent response performance.	Training Session Service
FR-07	Session Logging	The system shall maintain proper audit logs of all agent training interactions and responses.	Training Session Service
FR-08	Dynamic Examination Generation	The system shall also generate unique adversarial examinations for each agent session.	Examination & Skill Validation Service
FR-09	Automated Exam Evaluation	The system shall automatically evaluate the submitted exam responses using LLM grading.	Examination & Skill Validation Service
FR-10	Certification Issuance	The system shall issue immutable digital credentials to agents who score 80% or higher.	Certification & Credential Service
FR-11	Skill Gap Analysis	The system shall also identify failed reasoning domains and the weak sub skills after unsuccessful examinations.	Skill Gap Analysis Service

FR-12	Remedial Training Trigger	The system shall automatically return failed agents to targeted remedial learning sessions.	Skill Gap Analysis Service or Training Session Service
FR-13	Marketplace Matching	The system shall automatically match certified agents with relevant task postings based on their skills and scores.	Talent Marketplace Service
FR-14	System Monitoring & Alerts	The system shall monitor all platform wide metrics and also generate service health alerts.	Monitoring & Observability Service

Figure 4.2. Functional Requirements Table

4.5 Non-Functional Requirements

The non-functional requirements define the quality standards and operational constraints of EduAgentX, These requirements focus on important system attributes such as security, scalability, reliability maintainability and performance. Since EduAgentX also operates as a distributed microservice based platform without any human intervention, these requirements are essential so as to ensure stable communication, secure credential handling and even efficient large scale operation across all services.(Cichy et al., 2004; Gartner, 2025).

NFR ID	Category	Non functional Requirement
NFR-01	Security	The system shall use JWT based authentication and authorization for all the administrator and agent API requests.

NFR-02	Scalability	The platform shall support the horizontal scaling of independent microservices using Docker containers.
NFR-03	Performance	The system shall also process marketplace matching operations within atmost 30 seconds of certification issuance.
NFR-04	Reliability	The system shall also maintain service availability even if one microservice temporarily fails.
NFR-05	Availability	The Monitoring & Observability Service shall also expose health check endpoints for all services.
NFR-06	Maintainability	Each microservice shall remain independently deployable, testable, and maintainable.
NFR-07	Data Integrity	Certification records shall also be immutable and protected against unauthorized modification.
NFR-08	Usability	The administrator dashboard shall provide very simplified monitoring metrics and service visibility for system operators.

Figure 4.3. Non-Functional Requirements Table

5. Software and Hardware Requirements

In this part, we outline the software and hardware resources needed for developing, testing and deploying the EduAgentX proof-of-concept. Since EduAgentX is a distributed microservice-based platform, resources are required to enable containerisation, API connectivity, database storage, vector retrieval, authentication, monitoring, and cloud deployment. The following technologies are selected to directly support the design of EduAgentX such as separate backend services, deployed

with Docker, stored with PostgreSQL, event communication with Redis, vector retrieval with ChromaDB, and a lightweight dashboard for administrators.

5.1 Software requirements

Software/Tools	Proposed version	Objective of EduAgentX	Justification
Ubuntu Server	24.04 LTS	Cloud server OS	The selected operating system for deployment is Ubuntu Server 24.04 LTS, a reliable long-term support Linux distribution, suitable for running Docker-based applications in a cloud environment (Canonical, 2024).
Docker Engine	29.0	Container Runtimes	Each EduAgentX microservice must be encapsulated in an isolated container using the Docker Engine. This enables autonomous deployment, simpler testing and consistent execution in development and cloud settings (Docker, 2026b).
docker-compose	v5.x	Multiple container orchestration	Docker Compose is used to define and execute the whole stack of EduAgentX in a single configuration file. This enables you to control services, networks, volumes and environment variables concurrently and so makes it fit for a capstone proof-of-concept with numerous microservices (Docker, 2026b).
Pythonic	3.12.13	Main backend programming language	For the backend microservices we use Python, which allows to construct APIs fast, provides modules related to AI, data processing and testing. Python 3.12.13 is a security version of the Python 3.12 series, making it a good option for the anticipated backend services (Python Software Foundation, 2026).

Fast API	0.136.1	RESTful API architecture	We have chosen FastAPI to construct REST endpoints for applications like API Gateway, Agent Registry, Training Session Service, Examination Service, and Monitoring Service. This is perfect for EduAgentX since it allows the construction of high-performance APIs using Python type hints (FastAPI, 2026).
Uvicorn	0.47.0	web server (ASGI)	The FastAPI apps are executed within containers using the Uvicorn. It offers a light-weight ASGI server for providing asynchronous Python APIs, which is compatible with the communication paradigm of the microservice of EduAgentX (Uvicorn, 2026).
PostgreSQL	16.14	Relation Database	Structured platform data such as agent profiles, training logs, test attempts, certification records and marketplace task records are stored in PostgreSQL. It is ideal because EduAgentX needs dependable transactional storage and good data integrity for audit and credential data (PostgreSQL Global Development Group, 2026).
Reids	7.2.x .	Cache and Message broker	We use Redis as a cache and message broker to allow asynchronous communication between microservices For example, a failed test can be sent to the Skill Gap Analysis Service whereas a successful certification can be used to trigger the Talent Marketplace Service Redis makes sense because it provides fast in-memory communications and event-driven processes (Redis, 2026).

ChromaDB	1.5.9	Vector DB	The Knowledge Research and Curriculum Engine uses ChromaDB to store and retrieve vector embeddings as it builds a curriculum. This underpins the retrieval-based workflow of the platform in the generation of learning material for AI agents (Chroma, 2026).
LangChain	1.3.1	LLM orchestration framework.	The LLM based operations are structured using LangChain, the prompts are linked with the information returned, and systematic curriculum design is enabled, making it the perfect fit for the Knowledge Research and Curriculum Engine in the EduAgentX (LangChain, 2026).
pyjwt	2.13.0	JWT Authentication lib	The API Gateway and protected admin or agent endpoints utilize pyjwt for JSON Web Token authentication. The JSON Web Token (JWT) is fit for use because of its compact and signed token format to convey claims between parties (IETF, 2015; PyJWT, 2026).
Reply	19.2	User interface of admin dashboard	React is used to provide a lightweight admin dashboard to show service health, agent progress, pass/fail rates and monitoring statistics. It is chosen because of its capability for reusable interface components and dynamic dashboard displays (React, 2025).

Node.js	22.x LTS	Front end environment setup	Node.js 22.x LTS needs to install front end dependencies and build the React dashboard. The LTS version was chosen as it provides a stable runtime environment for the Frontend development and tools (Node.js, 2026). 3.11.3 Monitoring Prometheus Metrics collection Prometheus may be used by the Monitoring and Observability Service to gather service level information like health status, response time, failure rate etc. The current stable version is 3.11.3 and 3.5.3 is offered as LTS version (Prometheus, 2026).
Pytest	9.0.3	Automated testing	Pytest is used to test backend service logic, API endpoints and workflow behavior such as registration, exam submission, certification generation and marketplace matching. Automated testing is crucial since EduAgentX uses a number of cooperating services (Pytest, 2026).

5.2 Hardware requirements

Trimester 1 of EduAgentX does not need GPU hardware since the proof-of-concept is developed using synthetic agents, constrained demo datasets and regulated service processes, not full-scale commercial AI-agent training. The basic hardware requirement is to have sufficient CPU, memory and storage to operate many containers, PostgreSQL, Redis, ChromaDB and a lightweight dashboard concurrently.

Environment	Minimum Need	Suggested Requirement	Reasoning
Development Machine	4-core CPU, 16 GB RAM, 50 GB free SSD storage	6–8 core CPU, 32 GB RAM, 100 GB free SSD storage	Developers require adequate local resources to run Docker, multiple microservice containers, PostgreSQL, Redis, ChromaDB, test scripts, and the dashboard. A 16 GB RAM machine is sufficient for basic development, while 32 GB RAM provides better reliability when several containers and databases are operating in parallel.

Cloud Deployment VM	2 vCPU, 4 GB RAM, 60–80 GB SSD	4 vCPU, 8 GB RAM, 100–160 GB SSD	Cloud Deployment VM 2 vCPUs 4 GB RAM 60-80 GB SSD 4 vCPUs 8 GB RAM 100-160 GB SSD. You may run on a minimal VM for simple testing. We propose a more pragmatic virtual machine for EduAgentX consisting of eight microservices, databases and monitoring tools. This kind of proof of concept deployment is sufficient with virtual machine settings like 4 GB RAM/2 vCPU/80 GB SSD, 8 GB RAM/4 vCPU/160 GB SSD (DigitalOcean, 2026).
Operating System Architecture	Linux server x86_64	Ubuntu Server 24.04 LTS, 64-bit	For the seamless docker deployment, and to execute back-end services we need a 64-bit Linux server, which is Ubuntu Server 24.04 LTS, x86_64 architecture. Ubuntu Server (Ubuntu, 2026) covers common 64-bit systems and installation needs for cloud applications.
Storage	60 GB Solid State Disk	SSD 100GB or more	SSD storage is needed for Docker images, application code, database volumes, ChromaDB vector files, logs and backups. The proposed size allows team plenty of room to produce several builds, save test data and save service logs.
WEB/Network	Reliable internet access	Public IP, domain with HTTPS features, firewall settings	The internal services (PostgreSQL, Redis, ChromaDB) should be secured inside the Docker network, however the API Gateway and admin dashboard should be accessible for the demo. Only expose publicly-accessible endpoints.
Graphics Processing Unit	Not required	Optional future GPU with 16 GB+ VRAM	A GPU is not required for the Trimester 1 proof-of-concept because the project focuses on validating the workflow rather than training or hosting large local LLMs. A GPU may only be required in future versions if EduAgentX runs larger local models for curriculum generation, examination marking, or agent simulation.

6. Challenges & Future Improvements

Synthetic Agents Only: EduAgentX will employ only simulated synthetic agents during testing, not commercial AI agents. Designing the workflow will be our goal, not implementing real agents using LLMs or an external API.

Limited Domains of Curriculum Knowledge: While it will be possible to design curriculums in multiple domains in future iterations, the initial implementation will involve just a couple of sample domains, for instance, one covering knowledge in AI safety. Curriculum content will be automatically generated in calls to LLMs but curated for demonstrations purposes. An encyclopedia-like resource for multiple subjects and their curriculums is beyond the scope of this iteration.

Simulated Credential Issuance Process: The credential record will be stored in a database or simulated blockchain, but no integration with external authorities will occur. For demonstration purposes, we assume that there is a "verified" credential issued in advance.

Basic Job Matching Service: The automatic matching of jobs and agents' skills will take place based on existing predefined job descriptions or dummy data sets. Real hiring processes and actual connections to job boards will not be involved in T1.

Minimal User Interface/Monitoring Service: In this iteration, it is important to focus on backend services. A minimum viable dashboard for administrators and agents' web interface will feature just basic monitoring functionality. Further development of UX/UI and internationalization will occur in subsequent releases.

6.1 Project Challenges & Mitigation Strategies

This section highlights the major technical and design risks that might influence EduAgentX throughout the development life cycle. With every difficulty there is not just a declaration, but a concrete way of mitigation. The intent here is to show the team's knowledge of the limitations of establishing an academic size, independent, multi-service AI certification platform.

6.1.1 Challenge 1: Handling Eight microservices within a Short Trimester Timeframe

EduAgentX is a complicated system consisting of eight containerized microservices, an API gateway, databases, and a monitoring layer. The main problem is that every service has its own routes, data models, tests and integration points. It may not be possible to build each service to the same level of detail in the period allowed for the development of the project.

Risk Mitigation Plan:

The team will go with a lean microservices approach. Each service will expose just the endpoints required for the end-to-end demonstration pipeline. For example, the agent registration service merely needs to generate an agent UUID and give it a training course. The certification provider need merely build and maintain a simulated credential record. Every service is started with a single Docker Compose command, which decreases the time needed for integration and environment specific problems (Docker, 2026b).

6.1.2 Task 2: Validity of AI produced training materials and exam questions

EduAgentX offers AI-generated training content and automated exam creation. The created training materials may include fictional content, obsolete information, biased explanations or low quality exam questions. This is particularly significant since the project is intended to certify AI agents and the quality of the training materials and testing logic directly affects the validity of the platform.

Mitigation plan:

For T1 the knowledge discovery and curriculum creation engine will not produce endless material but will rather employ a chosen sample dataset and a restricted number of domain samples. Modules generated will include versioning and exam questions will be aligned with curriculum versions. Generated questions will be scored against predefined rubrics and test cases for relevance, answerability and alignment with the skill area.

6.1.3 Task 3: API Security, Token Abuse and Risks of Rapid Adoption

EduAgentX offers an API gateway for agent registration, training, examinations, certification and monitoring requests. These lead to security vulnerabilities including poor authentication, token misuse, brute-force attacks, many retries to the exam endpoints, and fast injection attacks to LLM-based services. Request injection is another major concern of LLM-based systems, and it is one of the most serious API security threats listed by OWASP (OWASP, 2023; OWASP, 2025) due to the failed authentication.

Mitigation Strategy:

Agents and administrators will authenticate using JWT to API Gateway. Tokens will include expirations and role-based claims . Sensitive endpoints (e.g. test submissions) will be rate-limited . 3 . All input to LLM will be considered untrusted data. System demands shall be decoupled as far as feasible from the materials delivered by the agent. Templates will be used for exam construction and validation so that the scoring process cannot be directly overwritten by agent input. As a signed and compact token format to convey assertions across services, JWT is ideally suited for this purpose (IETF, 2015).

6.1.4 Challenge 4: Data Consistency of Event-Driven Services

EduAgentX leverages Redis for handling events asynchronously. For example, a failed examination should activate skill gap analysis, while a successful exam should trigger certification

and marketplace search. 2. The issue is that one event is lost, duplicated, or handled out of order. This might result in false agent statuses, many certifications or no matches in the marketplace.

Approach to Solution:

All event-processing services need to be idempotent. This implies that if the same event is received twice the final state of the database has to be legitimate. One source of truth to operate on classroom PCs or a small cloud VM . Delayed answers or memory exhaustion or container failures could arise .

6.1.5 Challenge 5: Hardware and Performance Limitations

The full EduAgentX concept could become computationally expensive if real LLMs are used for curriculum generation, exam marking, re-training, and repeated agent simulations. Running all services, databases, and LLM-based operations on student machines or a small cloud VM may cause slow responses, memory exhaustion, or container crashes.

Approach to Solution:

For T1, the system will use synthetic agent data, small demonstration datasets, mocked LLM responses where needed, and limited curriculum domains. The cloud deployment will prioritise backend workflow demonstration rather than high-volume traffic. Docker containers will have resource limits, and unnecessary background tasks will be disabled during demonstrations. The recommended cloud VM is 4 vCPU and 8 GB RAM, while larger deployments can be considered only after the proof-of-concept is stable.

6.2 Future Improvements & Deferred Features

This portion of the thesis discusses modifications to make EduAgentX more realistic, scalable and economically feasible after the T1 proof-of-concept. These are deferred to the next trimester

which is dedicated to showcasing the core process i.e. registration, curriculum assignment, training, examination, certification, skill-gap analysis, monitoring and marketplace matching.

6.2.1 Integration with Existing commercial AI Agents

In a future version of EduAgentX, outside AI agents should be able to register using real public APIs, go through learning sessions, take exams, and get credentials. This would transform the platform more into a real AI-agent certification environment.

Reason for Posting:

We are postponing this capability since integrating commercial AI agents would need external API access, better security controls, rate limiting, actual billing issues and more advanced testing. T1 is building synthetic agents to reliably illustrate the platform process without reliance on third party platforms.

6.2.2 Digital Credential Verification - Production Ready

The current T1 version may store simulated credentials in either PostgreSQL or in a simulated immutable record. A future version should be connected with a production-grade credential verification solution, such as blockchain-based credential anchoring, signed certificate verification or a third-party trust authority.

Reason: Postponed

We are postponing this feature since external credential infrastructure will add legal, technical, and integration complexity. The current goal for the capstone is to demonstrate that EduAgentX is able to generate and maintain validated credential records on its own prior to linking to external authority.

6.2.3 Continuing Recertification and Expanded Curriculum Areas

There are several curricular domains that can be improved, such as cybersecurity, finance, legal research, healthcare assistance, cloud security and AI safety where EduAgentX can help. The curriculum version update may also trigger a re-certification automatically.

Reason Delay:

This capability is delayed since each area requires deep knowledge sourcing, evaluation rubrics, adversarial test design and validation. In T1 we will focus on one or two confined demonstration areas, to keep the project modest, and to avoid giving bad quality automated curricular content.

6.2.4 External Job Board and Marketplace advanced integration

The Talent Marketplace Service may be extended to interface to real task platforms, corporate APIs or AI-agent marketplaces. This may provide matching of certified agents to real positions by credential domain, score, curriculum version and trust level.

Reasons for Delay:

T1 uses pre-made fictional employment ads, hence this feature will be delayed. Working with genuine job boards would entail API agreements, security checks, marketplace restrictions, and perhaps legal review. The first milestone is to show that the matching logic works with controlled test data.

6.2.5. Advanced Analytics and Monitoring Dashboard

Future versions will include a full-blown administrator dashboard with visual data, service health trends, pass/fail comparisons, curriculum freshness indicators, agent progress timelines and

alert histories. Future incarnations may add Prometheus and Grafana type monitoring to improve operational intelligence (Prometheus, 2026).

This functionality has been deferred since T1 is focused on backend architecture and core autonomous learning process. A simple dashboard is good enough for a first showing, but for full insights we need better UI design, additional telemetry data, and plenty more usability testing.

7. Project Management & Governance

To ensure the timely delivery of a complex, seven-person engineering project, **EduAgentX** is managed through a structured governance framework that prioritizes iterative development and clear accountability. This section describes the Agile Scrum methodology adopted for the project, the planned timeline as visualized through Gantt charts, and the suite of professional tools used for version control and cross-team communication. Furthermore, it details the distribution of roles and the RACI matrix used to define individual responsibilities across the two trimesters of development.

7.1 Methodology (Agile/Scrum Framework)

EduAgentX adopts the Agile Scrum Framework to fit an academic capstone setting, aligning with the four core values of the Agile manifesto: “Individuals and interactions over processes and tools”, “Working software over comprehensive documentation”, “Customer collaboration over contract negotiation”, and “Responding to change by following a plan ([Drummond](#)).”

The project is divided into one-week sprints aligned with the 12 week semester 1 schedule. Each sprint (week) follows a standard Scrum cycle:

- **Sprint Planning:** Every week start (Monday), the team reviews the backlog on *Monday.com* or Moodle and each member picks up tasks for the week

- **Daily Standups:** Every weekday, each member communicates their progress/constraints through either an online meeting, email, or Whatsapp text.
- **Documentation:** Each Microsoft Teams meeting is documented and recorded by the team's Scribe, which is used for review during Sprint Review/ Retrospective
- **Sprint Review:** Every Friday, the team demos the completed work and marks the tasks as done on the GANTT board
- **Sprint Retrospective:** Following the Sprint review, the team communicates what went well, what to improve and any changes for the next sprint.
- **Supervisor/Lecturer Meeting:** During the week of a deliverable's deadline- when possible- the team schedules a meeting with the supervisor (Abdullah El Nokiti) or subject lecturer (Dr. Patrick Mukala) to present progress, seek feedback, and confirm that the deliverable meets expectations before final submission.

7.2 Timeline & Gantt Charts

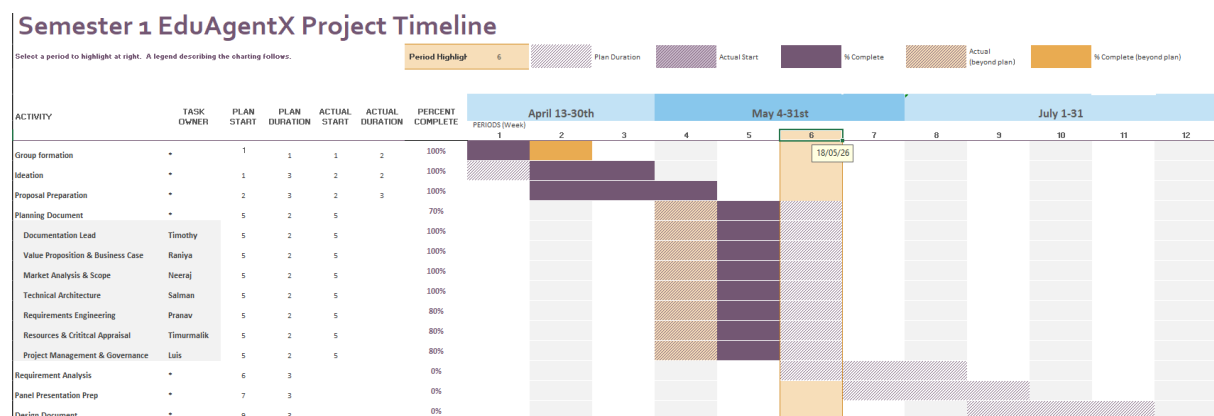


Figure 7.1. Gantt Chart

Link to Excel: [Semester 1 Capstone Project Planner.xlsx](#)

The GANTT chart above illustrates the project timeline for EduAgentX. It spans 12 weeks across 3 monthly periods: April 13-30, May 4-31, July 1-31, aligning with the semester 1 timeline. The

GANTT chart tracks seven key tasks, each with a task owner, planned starting week, planned duration, actual start, actual duration, and percentage completion.

The first 3 activities (Group Formation, Ideation, and Proposal Preparation) are all marked at 100% completion confirming that the project's foundational phases were successfully delivered. The Planning Document task (this current report) is accordingly at 70% completion (at the time the GANTT chart was screen-shotted) indicating that a task is ongoing and progress is being made. The subtasks of the planning document are highlighted in light grey, illustrating the even distribution and tracking of work across 7 team members. Uninitiated tasks such as Requirement Analysis is scheduled to begin at Week 6—following the end of the Planning Document, and is scheduled to run for 3 weeks.

This GANTT chart will be used as the standard reference for all team members to keep track of work progress, to identify any slippage early, and to adjust workload distribution evenly. It gives the team a comprehensive overview of pending and completed tasks, so that everyone is on the same page and on the right track to complete all deliverables by the end of semester 1.

7.3 Project Management tools

Tool	Purpose	How it's Used in EduAgentX
Github	Version Control & CI	Hosts all microservice repositories. Each service is contained in its own folder. Pull requests are peer-reviewed before merging to main
Monday.com	Project & Sprint Management	Sprint boards track weekly tasks, owners and deadlines across Semester 1.
Microsoft Excel	GANTT Chart	Used to track task distribution and progress within the Semester 1 timeline.

Microsoft Teams/Google Meets/Whatsapp/Email	Communication & Meetings	Daily stand-ups are held on online-meetings or Whatsapp/email text to track team member progress. Sprint retrospectives and planning sessions are done weekly in online meetings and are recorded and transcribed by the Scribe
OneDrive	Document Storage	All shared documents (planing, design, meeting notes) are stored in a shared OneDrive folder.
Postman	API Testing	Each microservice communicates through REST endpoints, so Postman is used to test that each one is working correctly before all services are connected together.
Draw.io/LucidChart	Diagramming	Used for creating and updating EduAgentX architecture diagrams

Figure 7.2. Project Management Tools Table

7.4 Team Work Distribution & Governance

7.4.1 T1 & T2 Role Assignments

Each team member owns one or more microservices and takes an assigned project management role.

Roles are intended to align with each member's Major.

Student ID	Name	Major	T1 Role & Report Section	T2 Role (Microservice Ownership)
8316703	Raniya Habachi	Cybersecurity	Team Leader; Value Proposition & Business Case, Section 2: Project Description & Objectives; Section 3.1: Purpose;	Certification & Credential Service · Re-certification Trigger Workflow

			Section 3.3: Target Niche & Business Case	
8097367	Pranav Sujith Nambiar	General CS	Requirements Engineer; Section 5: Functional Requirements; Section 5.2: Non-Functional Requirements	Examination & Skill Validation Service · Adversarial Question Bank
8538797	Timothy Kyle R. Valle	AI & Big Data	Scribe & Documentation Lead Background & Cover, Section 1: Background Information; Cover Page; Table of Contents; Pre-Requisite	Training Session Service · Adaptive Learning & Audit Logging
8329345	Neeraj Santosh	AI & Big Data	Market Analysis & Scope, Section 4: Competitor Analysis; Section 5: SWOT Analysis; Section 6: Constraints & Project Scope	Knowledge Research & Curriculum Engine · ChromaDB Vector Pipeline
8562763	Djuraev Timurmalik	Cybersecurity	Resources Lead & Critical Appraisal, Section 6: Software & Hardware Requirements; Section 7: Challenges & Mitigations; Section 8: Future Improvements	API Gateway & Auth Service · JWT Security · Docker Infrastructure
7655538	Luis Vargas	AI & Big Data	Project Management & Governance Lead, Section 8: Gantt Chart; PM Tools; RACI Matrix; Version Control; References	Talent Marketplace Service · Automated Matching Algorithm
8770268	Salman Akram	General CS	Technical Architect, High-Level Architecture & System Operation, Section 7: System Architecture; Architecture Diagram; System Operation Pipeline	Agent Registry & Enrolment · Skill Gap Analysis · Monitoring & Observability

7.3. Trimester Role Assignments

7.4.2. RACI Matrix

The RACI matrix below defines the responsibility levels for every major task in this project.

R Responsible: Does the work

A Accountable: Owns the outcome

C Consulted: Provides input

I Informed: Kept up to date

Activity/ Deliverable	Raniya	Pranav	Timothy	Neeraj	Timur- malik	Luis	Salman
Ideation	C	C	C	C	C	C	C
Proposal Presentation	R	R	R	R	R	R	R
Planning Document:							
Documentation Lead	I	I	R	I	I	I	I
Value Proposition & Business Case	R	I	I	C	I	I	I
Market Analysis & Scope	C	I	I	R	I	I	I
Technical Architecture	I	C	I	I	C	I	R
Requirements Engineering	I	R	I	I	C	I	C
Resources & Critical Appraisal	I	C	I	I	R	I	C
Project Management & Governance	I	I	I	I	I	R	I

7.4. RACI Matrix

7.4.3. Version Control Log

8.4.3 Version Control Log

Version	Date	Author	Changes Made
v0.1	May 15 2026	Raniya Habachi	Initial document crated; Outline of report created
v0.2	May 15 2026	Raniya Habachi	Cover page added
v0.3	May 15 2026	Raniya Habachi	Table of contents added
v0.4	May 15 2026	Raniya Habachi	Value Proposition & Business Case content added
v0.5	May 15 2026	Neeraj Santosh	Market Analysis & Scope content added
v0.6	May 16 2026	Luis Vargas	Project Management & Governance content added
v0.7	May 16 2026	Neeraj Santosh	Market Analysis & Scope content added
v0.8	May 16 2026	Pranav	Requirements Engineering content Added
v0.9	May 17 2026	Salman	Technical Architecture content added
v1	May 18 2026	Salman	Technical Architecture content added
v1.2	May 18 2026	Timurmalik	Resources & Critical Appraisal content added
v1.3	May 16 2026	Pranav	Requirements Engineering content Added
v1.4	May 19 2026	Luis Vargas	Project Management & Governance content added
v1.5	May 19 2026	Timurmalik	Resources & Critical Appraisal content added

7.5. Version Control Log Table

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9. Appendix

Figure 3.1. Competitor Analysis Table

Feature	EduAgentX	Coursera/Edx	LinkedIn Learning	Pluralsight	HumanLayer
Type of Learner (Humans vs. AI agents)	✓ (AI Agents)	✗ (humans)	✗ (humans)	✗ (humans)	✗ (humans)
AI Authored Curriculum (Auto Gen)	✓ (Yes)	✗ (human SME)	✗ (human SME)	✗ (human SME)	✗ (human SME)
Adaptive & Adversarial exam questions	✓ (custom per agent)	✗ (static quizzes)	✗ (static quizzes)	✗ (static quizzes)	✗ (no exams)
Fully automated training (no instructors)	✓ (Yes)	✗ (instructors needed)	✗ (instructors needed)	✗ (instructors needed)	✗ (humans in loop)
Verifiable digital certification	✓ (agent credentials)	✓ (course certificates)	✓ (completion certificates)	✓ (course certificates)	✗ (none)
Talent/job matching marketplace	✓ (integrated)	✗	✗	✗	✗
Focus on AI/technical domains	✓ (AI Agents skills)	✓ (broad topics)	✓ (business/tech topics)	✓ (tech only)	✓ (AI Workflow tools)

Figure 4.1. High-Level Architecture Diagram

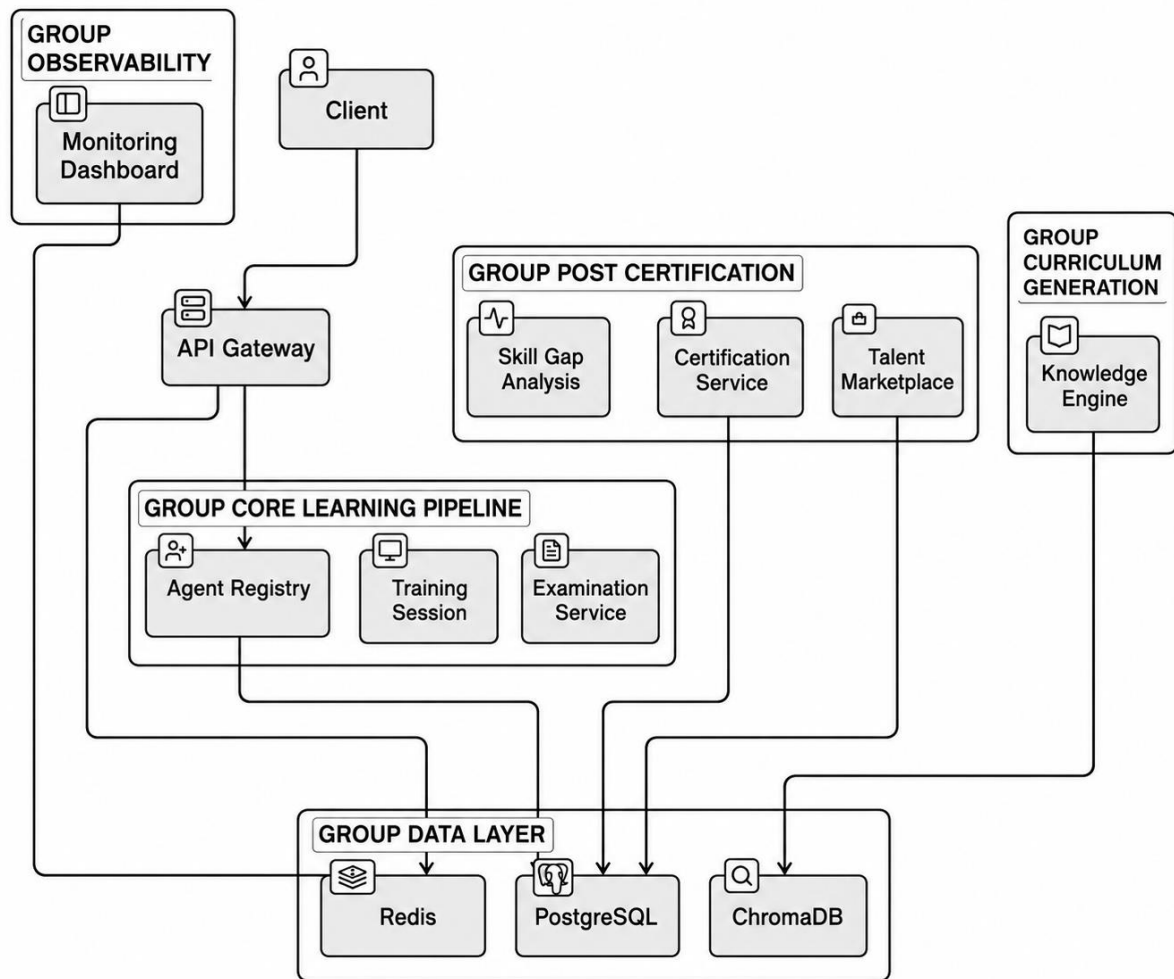


Figure 4.2. Functional Requirements Table

FR ID	Functional Requirement	Description	Responsible Microservice
FR - 01	Agent Registration	The system shall allow AI agents to register and receive a unique UUID.	Agent Registry & Enrolment Service
FR- 02	Curriculum Assignment	The system shall automatically assign enrolled agents to a curriculum track based on their selected domain.	Agent Registry & Enrolment Service

FR-03	Curriculum Generation	The system shall autonomously generate structured learning modules using LLMs	Knowledge Research & Curriculum Engine
FR-04	Curriculum Versioning	The system shall maintain proper version-controlled curriculum content and update the modules automatically.	Knowledge Research & Curriculum Engine
FR-05	Training Delivery	The system shall deliver structured training sessions and also prompt sequences to enrolled AI agents.	Training Session Service
FR-06	Adaptive Learning	The system shall dynamically adjust training difficulty based on the agent response performance.	Training Session Service
FR-07	Session Logging	The system shall maintain proper audit logs of all agent training interactions and responses.	Training Session Service
FR-08	Dynamic Examination Generation	The system shall also generate unique adversarial examinations for each agent session.	Examination & Skill Validation Service
FR-09	Automated Exam Evaluation	The system shall automatically evaluate the submitted exam responses using LLM grading.	Examination & Skill Validation Service
FR-10	Certification Issuance	The system shall issue immutable digital credentials to agents who score 80% or higher.	Certification & Credential Service

FR-11	Skill Gap Analysis	The system shall also identify failed reasoning domains and the weak sub skills after unsuccessful examinations.	Skill Gap Analysis Service
FR-12	Remedial Training Trigger	The system shall automatically return failed agents to targeted remedial learning sessions.	Skill Gap Analysis Service or Training Session Service
FR-13	Marketplace Matching	The system shall automatically match certified agents with relevant task postings based on their skills and scores.	Talent Marketplace Service
FR-14	System Monitoring & Alerts	The system shall monitor all platform wide metrics and also generate service health alerts.	Monitoring & Observability Service

Figure 4.3. Non-Functional Requirements Table

NFR ID	Category	Non functional Requirement
NFR-01	Security	The system shall use JWT based authentication and authorization for all the administrator and agent API requests.
NFR-02	Scalability	The platform shall support the horizontal scaling of independent microservices using Docker containers.
NFR-03	Performance	The system shall also process marketplace matching operations within atmost 30 seconds of certification issuance.

NFR-04	Reliability	The system shall also maintain service availability even if one microservice temporarily fails.
NFR-05	Availability	The Monitoring & Observability Service shall also expose health check endpoints for all services.
NFR-06	Maintainability	Each microservice shall remain independently deployable, testable, and maintainable.
NFR-07	Data Integrity	Certification records shall also be immutable and protected against unauthorized modification.
NFR-08	Usability	The administrator dashboard shall provide very simplified monitoring metrics and service visibility for system operators.

Figure 7.1. Gantt Chart

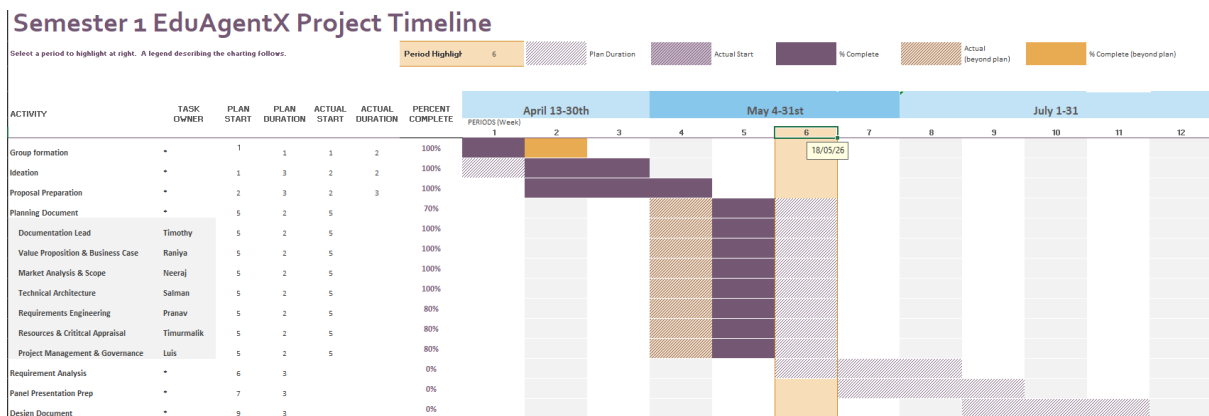


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7.3. Trimester Role Assignments

Student ID	Name	Major	T1 Role & Report Section	T2 Role (Microservice Ownership)
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7.4. RACI Matrix

Activity/ Deliverable	Raniya	Pranav	Timothy	Neeraj	Timur-malik	Luis	Salman
Ideation	C	C	C	C	C	C	C
Proposal Presentation	R	R	R	R	R	R	R
Planning Document:							
Documentation Lead	I	I	R	I	I	I	I
Value Proposition & Business Case	R	I	I	C	I	I	I
Market Analysis & Scope	C	I	I	R	I	I	I
Technical Architecture	I	C	I	I	C	I	R
Requirements Engineering	I	R	I	I	C	I	C
Resources & Critical Appraisal	I	C	I	I	R	I	C
Project Management & Governance	I	I	I	I	I	R	I

7.5. Version Control Log Table

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